

MA 101: Mathematics-I

Course Code	MA 101
Course Name	Mathematics-I
L-T-P-C	3-1-0-4
Prerequisites	NIL
One-time approval	5th Senate (Oct 2013)

Sem. Both

Course Contents

- **Functions of Several Variables:** Limit, continuity and differentiability of functions of two variables. Euler's Theorem, tangent plane and normal, change of variables, chain rule. Jacobians, Taylor's Theorem for two variables. Extrema of functions of two or more variables, Lagrange's method of undetermined multipliers;
- **Ordinary Differential Equations:** Solution of linear differential equations with constant coefficients, Euler-Cauchy Equations, Solution of second order differential equations by change of dependent and independent variables. Method of variation of parameters for second order differential equations. Numerical solution of ODE by Picard's method, Taylor's series, Euler method & Modified Euler method; Infinite Series: Convergence of infinite series, Comparison test, Ratio test, Root test, Raabe's test, Logarithmic test, Demorgan's test, Cauchy Integral test;
- **Solution in Series:** Solution in series of second order linear differential equations with polynomial coefficients. Bessel and Legendre equations and their series solutions. Properties of Bessel functions and Legendre polynomials;
- **Matrix Algebra:** Rank of a matrix, inverse of a matrix by elementary operations, Solution of linear simultaneous equations and their numerical solutions by Gauss Elimination and Gauss Seidel methods, eigen values and eigen vectors, Cayley-Hamilton theorem, diagonalization of matrices. Orthogonal, Hermetian, Skew-Hermetian, Normal and Unitary matrices and their elementary properties, quadratic forms.

References

1. Thomas G. and Finney R. L., **Calculus and Analytical Geometry**, 9th Edition, Addison Wesley.
2. Kreyszig E., **Advanced Engineering Mathematics**, 9th Edition, Wiley Eastern.
3. Grewal B. S., **Engineering Mathematics**, Khanna Publishers
4. Piaggio H. T. H., **An Elementary Treatise on Differential Equations and their Applications**, G. Bells & Sons Ltd.
5. Simmons G. F., **Differential Equations**, TMH Edition
6. Prasad C., **Mathematics For Engineers**, 19th Edition, Prasad Mudralaya.
7. Gerald C. F. and Wheatley P. O., **Applied Numerical Analysis**, 6th Edition, Wesley.
8. Krishnamurthy E. V. and Sen S. K., **Applied Numerical Analysis**, East West Publication.